Çağrı Şahin
Department of Software Engineering
Muş Alparslan University
49250 Muş, Türkiye

November 7, 2026

Editor-in-Chief
IEEE Access
IEEE Publishing Operations

Dear Editor,

We submit the manuscript "Less a Weak Source than a Different One: Vulnerability, Similarity and Direction in CNN-ViT Adversarial Transfer under an Architecture-by-Initialization Design" for consideration as a Regular Article in *IEEE Access*.

In our adversarially trained ResNet-18 and ViT-Tiny pair, examples crafted on the ViT fool the CNN less often than the reverse, so the ViT looks like a weak transfer source; transfer studies disagree on the direction of this asymmetry. With one CNN and one ViT this asymmetry cannot be divided into a source part and a target part. The manuscript factors each log target-correct transfer rate into target vulnerability, a symmetric pairwise similarity and a directional residual, averaged over a fixed panel of seven robust CNNs from RobustBench. On CIFAR-10, a share of $0.936\ [0.925, 0.948]$ (90% intervals over training runs, conditional on the test images and the panel) of ViT-Tiny's panel source gap relative to ResNet-18 lies in the symmetric term, where a pure source effect would give one half, and the ViT's directional term ($+0.012$) is equivalent to zero within the registered bound of ± 0.10. Most of what looks like a weak source is dissimilarity that runs in both directions.

The study was designed so that this reading could fail. Six new runs complete a two-by-two design crossing architecture with initialization on a matched pipeline, and its predictions were registered on 29 October 2026 before any of these runs; an addendum covering CIFAR-100, ImageNetV2 and gradient alignment was registered on 30 October 2026 before any of its matrices was computed. Trained from scratch, the ViT is less similar to the panel than a ResNet-18 ($-0.364\ [-0.415, -0.314]$), a contrast the design cannot separate from the ViT's lower clean accuracy; ImageNet initialization narrows it more for the ViT ($+0.260$), and averaged over initialization the architecture leaves the directional term within the bound ($+0.003$). The registered tests ran through the frozen analysis script or, for the addendum, through a wrapper around it and a gradient script whose digests were recorded before their first run; every analysis output stores the digests of the scripts that produced it. The manuscript also lists what the registration does not cover, among them a first CIFAR-10 seed seen in advance, a refutation rule widened to both signs after data had been seen, and one sensitivity analysis computed after the results.

The abstract, introduction and conclusion bound the claim in the same terms. A small directional term holds only on CIFAR-10, on the log scale and against a CNN panel: on CIFAR-100 the symmetric share replicates (0.840) but the registered equivalence of the ViT's directional term is refuted ($+0.143$), and on the logit scale that term is strongly negative. The ImageNetV2 evidence is descriptive and gradient alignment is reported as an association, not a mechanism. The decomposition

itself is an accounting identity close to Caussinus's quasi-symmetry model; its content lies in the values its terms take.

A preliminary version, limited to CIFAR-10 and to the same pair, has been submitted to ATEEC 2026 and is under review; the manuscript cites it and describes the overlap in its introduction. That version reports the conditional decomposition and the four-protocol comparison, while the panel decomposition, the two-by-two design, the external panels and the registration are new. The manuscript also corrects one claim of the conference version: the direction of the asymmetry is not stable once the two architectures are matched in clean accuracy.

As stated in the Acknowledgment, the authors used Claude (Anthropic) to assist with drafting and revising the text and with implementing the analysis and verification scripts; the study design, the protocol and all conclusions are the authors', who verified every reported number against the analysis records. No figure is produced by an image-generation model.

Apart from the conference version described above, the manuscript has not been published or submitted elsewhere. Both authors have approved the submitted version and declare no conflict of interest. The registration documents, the frozen analysis script and the cell tables are available, with the analysis pipeline and per-sample records, in the public repository at `https://github.com/cagrisahin58/cnn-vit-robustness-protocols`; they are also time-stamped in the development repository (commits e32f2ce and 7325419).

Thank you for considering our manuscript.

Sincerely,

Çağrı Şahin
Department of Software Engineering
Muş Alparslan University, 49250
Muş, Türkiye
Email: c.sahin@alparslan.edu.tr